Table 6.1

GANTT CHART

Task ID	Task Name	Task Lead	Start Date	End Date	JUN 2026				JULY 2026				AUG 2026				SEPT 2026				OCT 2026				NOV 2026			
					W 1	W 2	W 3	W4	W 1	W 2	W 3	W 4	W 1	W 2	W 3	W 4	W 1	W 2	W 3	W 4	W 1	W 2	W 3	W 4	W 1	W 2	W 3	W 4
1	**INTRODUCTION**																											
1.1	Create an introduction	Basubas	June 25	June 27																								
1.2	Explain the problem, research gap and solution	Isagunde	June 25	June 27																								
1.3	Write the specific objectives	Basubas	June 27	June 28																								
1.4	Define what the system includes and excludes	Isagunde	June 27	June 28																								
1.5	Explain who benefits from the study	Catarig	June 25	June 28																								
1.6	Show the study process	Catarig	June 28	June 29																								
1.7	Define the definitions of terms	Borces	June 25	June 29																								
1.8	Make the questionnaire	Basubas	June 25	July 1																								
1.9	Collect responses	Isagunde	June 25	July 1																								
1.10	Add citations	Basubas	June 30	July 1																								
1.11	Revise Chapter 1	All	June 30	July 1																								
2	**REVIEW OF RELATED LITERATURE AND STUDIES**																											
2.1	Identify and explain the theories	Catarig	July 2	July 4																								
2.2	Research journals	Borces	July 2	July 5																								
2.3	Gather and summarize literature	Isagunde	July 2	July 5																								
2.4	Review related studies	Basubas	July 2	July 5																								
2.5	Compare DriveAlert with existing systems and identify the research gap	Isagunde	July 2	July 6																								
2.6	Add citations, organize references, and revise Chapter 2	Basubas	July 6	July 8																								
3	**RESEARCH METHODOLOGY**																											
3.1	Select and explain the Agile methodology	Isagunde	July 4	July 5																								
3.2	Define the system requirements and development plan	Basubas	July 4	July 7																								
3.3	Prepare the Business Model Canvas	Basubas	July 4	July 8																								
3.4	Create the Driver and Trusted Contact program workflows	Borces	July 5	July 7																								
3.5	Complete the Validation Boards for Stages 1 and 2	Basubas	July 7	July 9																								
3.6	Prepare the Gantt charts	Isagunde	July 6	July 10																								
3.7	Create the Functional Decomposition Diagram	Catarig	July 4	July 6																								
3.8	Create the Use Case Diagram	Basubas	July 9	July 10																								

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3.9	Design the mobile and web user interfaces	Borces	July 5	July 7																								
3.10	Create the storyboard	Basubas	July 9	July 10																								
3.11	Identify the required database entities, attributes, and relationships	Borces	July 6	July 9																								
3.12	Create the Entity Relationship Diagram	Basubas	July 7	July 10																								
3.13	Prepare the Data Dictionary	Catarig	July 7	July 9																								
3.14	Create the Network Design	Isagunde	July 6	July 10																								

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					W 1	W 2	W 3	W4	W 1	W 2	W 3	W 4	W 1	W 2	W 3	W 4	W 1	W 2	W 3	W 4	W 1	W 2	W 3	W 4	W 1	W 2	W 3	W 4
3.15	Create the Network Model	Isagunde	July 6	July 10																								
3.16	Describe the system development	Basubas	July 10	July 11																								
3.17	Create the Technology Stack Diagram	Basubas	July 10	July 11																								
3.18	Define the software and hardware specifications	Borces	July 10	July 11																								
3.19	Prepare the Program Specification	Catarig	July 10	July 11																								
3.20	Prepare the Team Composition	Isagunde	July 10	July 10																								
3.21	Prepare the Appendices	Basubas	July 12	July 14																								
3.22	Prepare the List of Modules	Basubas	July 12	July 13																								
3.23	Revise the Gantt Chart and Agile Sprint Plan	Isagunde	July 13	July 14																								
3.24	Revise Chapter 3	All	July 12	July 15																								
3.25	Finalize the manuscript	All	July 16	July 22																								
4	**DEVELOPMENT TRANSITION PREPARATION**																											
4.1	Review the finalized manuscript requirements, diagrams, and module boundaries	All	July 18	July 20																								
4.2	Prepare development tools, accounts, hardware requirements, and project resources	Basubas and Borces	July 18	July 22																								
4.3	Assign module ownership, repository workflow, and coding standards	All	July 20	July 22																								
5	**PROJECT FOUNDATION, IOT SOURCING, AND MANUSCRIPT COMPLIANCE**																											
5.1	Create the Git repository and configure Android, web, and Firebase projects	Basubas, Isagunde, and Borces	July 23	July 28																								
5.2	Refine the local and cloud database schema and account structures	Borces	July 23	July 29																								
5.3	Identify suppliers and source the required IoT components	Basubas and All	July 23	August 3																								
5.4	Review component specifications, compatibility, and proposed wiring	Basubas and Borces	July 25	August 4																								
5.5	Establish local storage and initial database operations	Borces	July 27	August 4																								
5.6	Prepare project structure and initial account-screen layouts	Isagunde and Catarig	July 27	August 4																								
5.7	Revise the manuscript and diagrams based on review feedback	All	July 23	August 4																								
5.8	Prepare compliance documents and route the approval sheet	All	July 27	August 5																								
5.9	Check builds and documents, commit and push changes, and review readiness	All	August 3	August 5																								
6	**IOT ASSEMBLY, COMMUNICATION, AND CORE APPLICATION FUNCTIONS**																											
6.1	Assemble the camera, speaker, power connections, and mounting	Basubas and All	August 6	August 9																								
6.2	Configure camera firmware, capture settings, and device services	Basubas	August 8	August 11																								
6.3	Implement Bluetooth provisioning and local Wi-Fi connection	Basubas	August 9	August 14																								
6.4	Implement the IoT HTTP stream and Android video receiving	Basubas and Borces	August 10	August 16																								

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6.5	Verify onboard speaker playback and initial warning commands	Basubas and Catarig	August 11	August 16																								
6.6	Develop Android account functions, setup screens, and navigation	Basubas and Isagunde	August 6	August 16																								
6.7	Develop web authentication, routing, and initial account views	Isagunde and Catarig	August 6	August 16																								
6.8	Test power, streaming, reconnection, and account functions	All	August 14	August 19																								
6.9	Upload tested firmware, push changes to GitHub, and review connection readiness	Basubas and Borces	August 18	August 19																								
7	**MEDIAPIPE INTEGRATION AND LIVE FACIAL MEASUREMENTS**																											
7.1	Integrate MediaPipe Face Landmarker into Android processing	Basubas	August 20	August 25																								
7.2	Build live frame processing from the IoT camera stream	Basubas and Borces	August 22	August 28																								
7.3	Identify eye and mouth landmarks and calculate EAR and MAR	Basubas and Borces	August 23	August 30																								
7.4	Calculate head-pitch measurements from facial transformations	Basubas	August 24	August 30																								
7.5	Implement Driver-area selection and tracking continuity	Basubas	August 25	September 1																								
7.6	Implement facial-lighting and visibility checks	Basubas and Borces	August 26	September 1																								
7.7	Develop camera-alignment and facial-visibility feedback	Isagunde	August 27	September 2																								
7.8	Test landmark stability, lighting, delay, and Android compatibility	All	August 29	September 2																								
7.9	Commit and push processing changes and review calibration readiness	Basubas and Borces	September 1	September 2																								
8	**CALIBRATION, DROWSINESS DETECTION, AND WARNING CONTROL**																											
8.1	Implement guided Driver calibration and stored calibration values	Basubas	September 3	September 8																								
8.2	Implement prolonged-eye-closure detection using calibrated EAR	Basubas	September 4	September 10																								
8.3	Implement yawning detection using calibrated MAR	Catarig and Basubas	September 4	September 10																								
8.4	Implement relative head-pitch and temporal head-down detection	Basubas	September 5	September 11																								
8.5	Implement confirmation, episode grouping, and warning-stage rules	Basubas	September 8	September 13																								
8.6	Connect staged warnings to the onboard speaker command	Basubas and Isagunde	September 9	September 14																								
8.7	Implement local monitoring-session and alert-history records	Borces	September 9	September 15																								
8.8	Test calibration, false alerts, stage progression, and offline records	All	September 12	September 16																								
8.9	Commit and push tested changes and review detection reliability	All	September 15	September 16																								
9	**TRUSTED CONTACT SHARING AND SYSTEM INTEGRATION**																											
9.1	Implement connection requests, approval, and Driver relationships	Borces and Catarig	September 17	September 22																								

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9.2	Implement approved access, revocation, and security rules	Borces	September 18	September 24																								

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9.3	Create eligible Stage 3 transition and persistence records	Basubas and Borces	September 19	September 25																								
9.4	Implement synchronization queues, retries, and status updates	Basubas and Borces	September 21	September 28																								
9.5	Develop mobile shared-record views and Driver alert summaries	Isagunde and Catarig	September 20	September 27																								
9.6	Connect the web portal to approved Driver records and requests	Isagunde and Catarig	September 20	September 27																								
9.7	Integrate mobile, web, IoT warnings, records, and synchronization	All	September 23	September 29																								
9.8	Test sharing eligibility, revocation, offline queues, and full workflow	All	September 26	September 30																								
9.9	Push tested changes, deploy the web test build, and review integration	Isagunde and Borces	September 29	September 30																								
10	**INTEGRATED TESTING AND SYSTEM REFINEMENT**																											
10.1	Test mobile and web accounts, setup, validation, and navigation	Borces and Catarig	October 1	October 4																								
10.2	Test camera tracking, lighting pauses, and connection recovery	Basubas and Isagunde	October 2	October 7																								
10.3	Test calibration, sign confirmation, warning stages, and speaker response	Basubas and Catarig	October 3	October 8																								
10.4	Test local records, alert summaries, settings, and retention	Isagunde and Borces	October 3	October 9																								
10.5	Test approved sharing, revocation, synchronization, and retries	Borces and Catarig	October 5	October 10																								
10.6	Test web responsiveness, usability, and device compatibility	Isagunde and Catarig	October 5	October 11																								
10.7	Correct integrated defects and perform focused regression tests	All	October 8	October 14																								
10.8	Update manuscript descriptions and diagrams to match the system	All	October 8	October 14																								
10.9	Push fixes, generate a test APK, and update the web test deployment	Basubas and Isagunde	October 12	October 14																								
11	**CONTROLLED EVALUATION AND RELEASE PREPARATION**																											
11.1	Finalize controlled-evaluation procedures, forms, and device setup	All	October 15	October 17																								
11.2	Conduct controlled eye-closure, yawning, and head-down trials	All	October 17	October 22																								
11.3	Evaluate audible warnings, offline monitoring, and approved sharing	All	October 18	October 23																								
11.4	Record false alerts, missed detections, delays, and test results	Basubas and Borces	October 18	October 24																								
11.5	Resolve remaining mobile, web, IoT, and synchronization defects	All	October 21	October 26																								
11.6	Analyze evaluation findings against the study objectives	Basubas and Catarig	October 23	October 27																								
11.7	Complete manuscript findings, diagrams, and supporting evidence	Isagunde and Borces	October 24	October 28																								
11.8	Run complete regression tests and verify release readiness	All	October 25	October 28																								
11.9	Push verified changes, generate the release-candidate APK, and update the web	Basubas and Isagunde	October 27	October 28																								
12	**FINAL SUBMISSION, DEPLOYMENT, AND PRESENTATION**																											

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12.1	Finalize manuscript formatting, references, and required corrections	All	October 29	November 1																								
12.2	Complete approval requirements, compliance documents, and appendices	All	October 29	November 3																								
12.3	Verify manuscript completeness and prepare the submission package	All	November 1	November 3																								
12.4	Submit the final manuscript	All	November 4	November 4																								
12.5	Resolve final system issues and verify corrected functions	All	October 29	November 7																								
12.6	Generate and install the final APK and upload verified IoT firmware	Basubas and Borces	November 5	November 9																								
12.7	Deploy and verify the Trusted Contact web portal and access rules	Isagunde and Borces	November 5	November 9																								
12.8	Prepare presentation materials and rehearse the demonstration	All	October 29	November 10																								
12.9	Push final changes and verify deployment, presentation, and project completion	All	November 9	November 11																								

LEGEND	COMPLETE
	ONGOING
	INCOMPLETE

This section displays tasks and their associated timeframes in a horizontal bar format, enabling researchers to organize, track, and coordinate activities efficiently. It provides a comprehensive overview of the project's progress, improves communication, and ensures that project requirements are completed on schedule. Using the Agile Methodology, the Gantt chart is helpful due to the methodology's iterative nature, allowing each phase to be clearly represented and tracked